Quest 8: The Entgineering

Lord of the Wings · shadows on a hexagonal field, in six directions, while the Ents grow

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION

Energy sweeps across a hexagonal field of Ents (heights 1–9) in one direction; each Ent casts a shadow that drops by 1 per hex, and you count the Ents that stick out above the shadow reaching them.

INPUT

A hex grid drawn as text: digits are Ents, dots only keep the layout.

OUTPUT

The number of Ents that receive energy (one unit each).

WHAT CHANGES BETWEEN PARTS

Part I uses one direction (↘). Part II adds up all six hex directions on the same field. Part III makes the field change: lit Ents grow, height-9 Ents leave as warriors and come back as seeds, and you run the six-direction cycle until 100,000 warriors have left.

The story

With Sarumonk gone and the team out of the cage, the poisoned food is still a problem, so the hero turns to the Ents, who don't eat bread anyway. They offer to grow an army of wooden warriors on hexagonal growing fields, but first they need to know how much energy a field uses: a single test round, then a full circuit of the moving energy source, and finally the whole growing process.

Understanding the question

Reading the hex grid

The field is a hexagon squeezed into a text grid. Ents sit on every other character, and each row is shifted by one, so the six neighbours of the Ent at (row *r*, column *c*) are:

Direction	↘	↙	←	↖	↗	→
Next hex	(r+1, c+1)	(r+1, c−1)	(r, c−2)	(r−1, c−1)	(r−1, c+1)	(r, c+2)

Left and right move **two** columns, because the character in between is a layout dot. The diagonals move one row and one column.

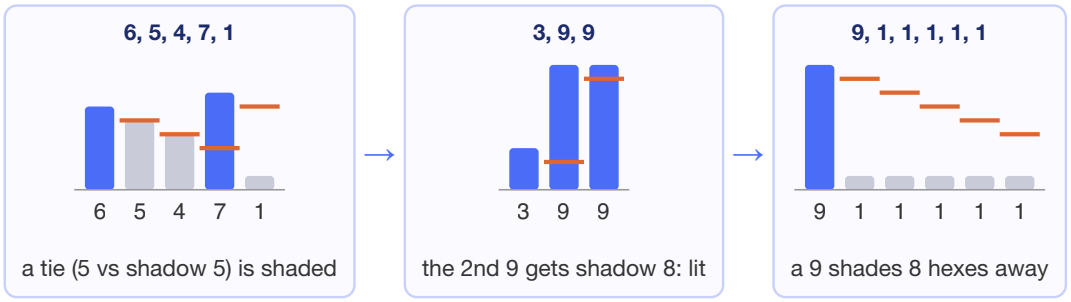
Shadows along one line

Energy travels in straight lines in the given direction. Walk along one line, from where the energy enters:

- An Ent of height h casts a shadow of $h - 1$ on the next hex, $h - 2$ on the one after, and so on.
- Every Ent casts its shadow, **even if it is itself shaded**. Overlapping shadows don't add up: the highest one wins.
- An Ent receives energy only if it is **taller** than the shadow reaching it. Equal height means shaded.

So a useful running value is the **"skyline"** S at each hex: the higher of the Ent's own height and the shadow arriving there. The shadow arriving at the next hex is $S - 1$. An Ent of height h is lit exactly when $h > S_{prev} - 1$, i.e. when $h \geq S_{prev}$.

Extra examples of my own (one line each, worked by hand)



Blue = lit, grey = shaded; the orange tick is the shadow reaching that hex (one less than the tallest of Ent-or-shadow before it).

And a whole small field (5 rows, 14 Ents), run through a copy of my Part I solution. The lit ones are bold, the shaded ones are struck through:



before ↘: 10 Ents lit

For example, in the ↘ line 2 → 1 → 9 (rows 2–4) the 1 is shaded by a shadow of exactly 1, and the 9 behind it is lit. In the line 5 → 9 → 4 → 3 (rows 0–3), the 9 shades the 4 (shadow 8) and the 3 (shadow 7). Running the same field through a copy of my Part II solution gives 10, 8, 10, 7, 10 and 9 for the six directions: **54** in total.

Shared approach: reading the field

The field is kept exactly as drawn: a grid of characters where digits are Ents and dots are layout. Neighbours are found with the row/column steps in the table above.

PART I One test round (↘)

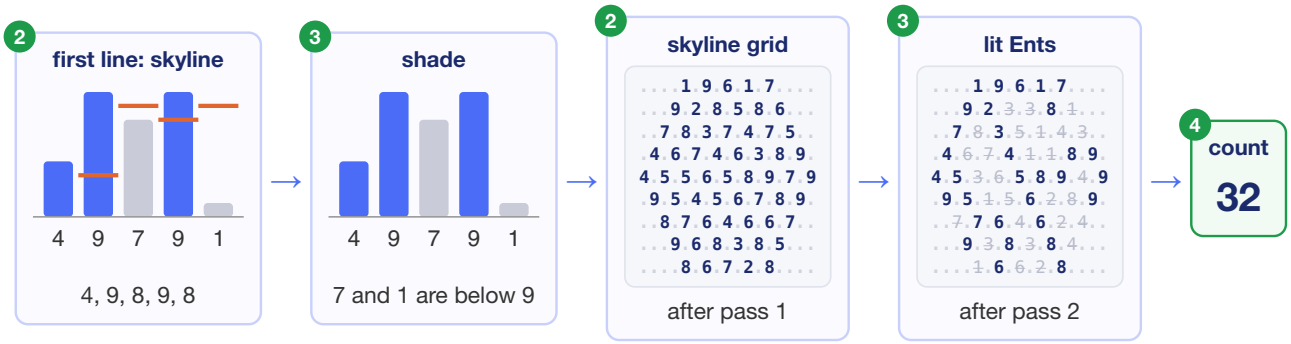
Energy comes from the upper left and travels ↘. Count the Ents that receive it.

Step-by-step solution

1. Read the field as a grid.
2. Go through the grid top to bottom, left to right. For every Ent, look at the hex up-left of it: if there is none, its skyline is its own height; otherwise its skyline is the higher of its own height and (that hex's skyline – 1).
3. Go through again: an Ent whose height is below the up-left hex's skyline is shaded.
4. Count the Ents that are not shaded.

How my solution works

Two passes, top to bottom: first work out every hex's **skyline** (the higher of its Ent and the shadow arriving), then shade each Ent that is below the skyline of the hex before it. The first ↘ line of the sample, then the whole field:



Dots match the step numbers above. Lit Ents are bold, shaded ones struck through.

> **Two passes are needed:** a shaded Ent still casts a shadow, so the skyline must be finished before any Ent is removed.

> **A small thing:** a dot in front of an Ent is treated as "the line starts here". That's right for a solid hexagon, but if a field had an empty hex *inside* it, a shadow ought to pass over the gap (dropping by 1) instead of stopping. Your inputs have no such gaps (checked), so this never matters.

PART II A full circuit of six directions

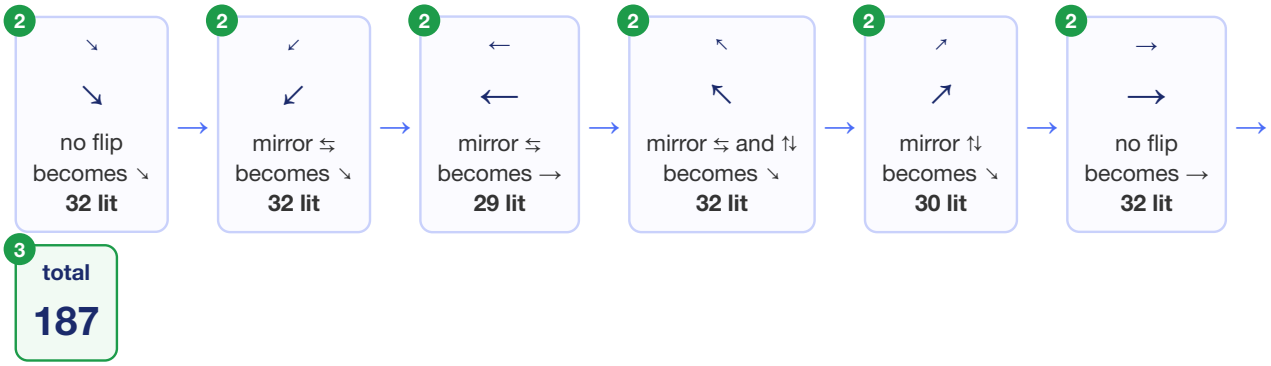
The same field is lit from all six directions in turn (↘, ↙, ←, ↖, ↗, →). Add up the lit Ents of the six rounds.

Step-by-step solution

1. Turn Part I into a reusable step that takes a look-back step: (1 row, 1 column) for diagonals, (0 rows, 2 columns) for horizontals.
2. For each of the six directions, mirror a fresh copy of the field so that the direction becomes ↘ or →.
3. Run the step on each copy and add the six counts.

How my solution works

Mirroring the field doesn't change which Ents are lit. So one shadow rule handles everything: mirror a **fresh copy** of the field until the direction becomes ↘ (look back 1 row, 1 column) or → (look back 0 rows, 2 columns), then count:



Dots match the step numbers above. ⇆ = reverse each row, ⇅ = reverse the row order. On my small field from the intro the six rounds give 10 + 8 + 10 + 7 + 10 + 9 = 54.

> **Why two settings:** a 60° turn isn't a simple grid operation on a hex grid stored as text, but mirrors are. Under mirroring the six directions fall into two groups: the four diagonals and the two horizontals.

PART III The growing field

Now the field changes after every round, and the source keeps cycling ↘, ↙, ←, ↖, ↗, →, ↘, …

1. First, all shadows for the round are worked out on the field as it is.
2. Every lit Ent uses one unit of energy and grows by 1.
3. A lit Ent of height 9 becomes a warrior instead. It leaves a **seed**, which stays underground for the next 3 rounds (no energy, no shadow) and sprouts as a height-1 Ent at the start of the round after that.
4. Stop after the round in which the total number of warriors first reaches 100,000, and report the total energy of all rounds.

A seed has no height, but shadows still **pass over** it and keep shrinking by 1 per hex, as over any other hex.

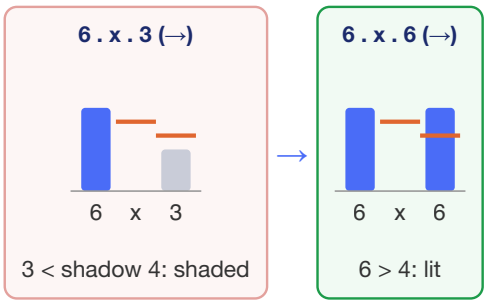
Worked example (sample)

The puzzle shows the first ten rounds; my solution (instrumented) gives exactly the same numbers:

Round	1	2	3	4	5	6	7	8	9	10
Direction	↘	↙	←	↖	↗	→	↘	↙	←	↖
Energy	32	31	29	22	25	27	28	27	31	26
New warriors	8	6	5	4	5	1	8	1	6	1
Energy so far	32	63	92	114	139	166	194	221	252	278

The army reaches 100,000 warriors in round **32008**, and the total energy is **899889**, both matching the puzzle.

Seeds and shadows



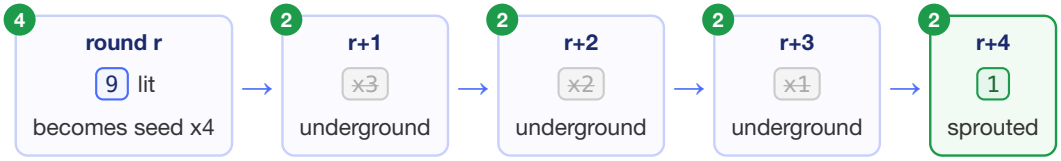
x = a seed: height 0, never lit, but the shadow passes over it.

Step-by-step solution

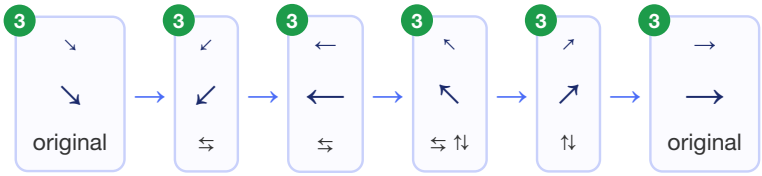
1. Keep the field between rounds; store seeds as countdowns `x4` … `x1`.
2. At the start of each round, count every seed down by one, turning it into a height-1 Ent when its countdown runs out.
3. Flip the field into the right orientation for this round's direction and count the lit Ents, treating seeds as height 0 and never lit.
4. Grow every lit Ent by 1; a lit 9 becomes a new warrior and a seed `x4`.
5. Add the round's energy and warriors; repeat until there are at least 100,000 warriors, then print the total energy.

How my solution works

The shadow step is Part II's, with seeds treated as height 0 (never lit). Each round: **sprout** (count seeds down), **orient** the field, **light** it, then **grow** every lit Ent (a lit 9 becomes a warrior and a seed). A seed's life:



Starting at x4 works because sprouting happens *before* each round. Unlike Part II, the flips are applied to the field itself and **stay**, each round flipping on from the last:



Dots match the step numbers above. Under each direction: the field's orientation in that round. After →, it's back to the original for ↘.

Fragile: the orientation is carried from round to round. This is correct, and the sample matches all ten rounds and the final 899889. But each case only works because the previous case left the field in a specific flip, so the six cases can't be reordered, skipped or tested on their own. If the rounds ever started at a different direction, or one case were changed, every later round would be lit from the wrong side, and the totals could still look plausible. Part II's approach (flip a fresh copy from the original each time) avoids this, at the cost of flipping the growth back afterwards.

Cost: 32008 rounds, each making a few full copies of the grid; the whole run takes about half a second on the sample and on the real input.

Summary

Part	Directions	Field changes?	Key trick in my solution	Sample
I	↘ only	no	a skyline grid (Ent or shadow, whichever is higher), then compare each Ent to the previous skyline	32
II	all six, once	no	one shadow rule with a look-back step (1, 1) or (0, 2), and mirrored copies of the field	187
III	all six, cycling	yes: growth, warriors, 3-round seeds	seeds as countdowns, seeds at height 0, and flips carried between rounds	899889

Key ideas: a shadow that drops by 1 per step is a **running maximum** ("skyline") along each line; a hex grid stored as text has two kinds of straight lines (diagonals and horizontals), and **mirroring** turns every direction into one of them; and in a simulation, decide everything for the round first (shadows), then apply the changes (growth).